



Experience

Production Engineer — Meta, Traffic Foundation

 Menlo Park, CA, USA  Dec 2022 – Present

- Own and develop observability services for Meta's global CDN infrastructure serving billions of users:
 - *Synthetic End-to-End Monitoring (Synthetic E2E)*: High-sensitivity synthetic availability probing for surfacing issues across load balancers and networking devices (sub-0.001% error rate detection, <1 min signal latency, covers STUN/TURN, QUIC, HTTP/1-3, and proprietary protocols). Implemented in Rust with Python-based config generation.
 - *Traceroute Service*: Periodic and on-demand traceroute collection for localizing external routing and peering failures to individual ASNs, routers, and peering points. Implemented in Rust with Python-based config generation.
 - *Investigation Tooling*: Tool suite for correlating monitoring signals across services and localizing faults to individual clusters, hosts, and processes. Implemented in Hack and Python.
- Investigated and reproduced a complex BGP misconfiguration causing near-total IPv6 loss across 2% of fleet; drove implementation of prevention and detection measures such as prefix filtering and cluster-level health measurement.
- Presented BGP reproduction investigation as an internal conference talk to ~100 infrastructure and platform engineers across diverse teams and fields.
- Rewrote Synthetic E2E from Python to Rust, reducing memory footprint and freeing over \$20,000 of capacity.
- Rewrote Traceroute Service from C++ to Rust after team ownership transfer using AI assistance; validated continued correctness using existing unit tests and new integration tests.
- Authored 15+ concise RFCs and long-term roadmaps establishing directional clarity and priority justifications, driving team planning and alignment during an extended absence.

Production Engineering Intern — Meta, Video CDN

 Menlo Park, CA, USA  Jun 2021 – Sep 2021

- Designed and implemented a Python-based automated canary tool for Edgetee, a video streaming service, improving developer experience and deployment confidence.
- Cut Edgetee binary canary build times by over 50% by implementing parallel CI job execution.

Projects

Caligula Disk Burner

 github.com/ifd3f/caligula ★ ~2.1k stars  Mar 2023 – Present

A Rust-based disk imaging tool focused on usability and performance, widely used by embedded systems developers, infrastructure engineers, and Linux power users.

- Engineered a multithreaded I/O pipeline using custom double-buffered SPSC channels, reducing hashing thread futex time from 30% to 3%.
- Built a reproducible CI system using end-to-end NixOS VM tests to catch regressions in correctness and UI.
- Implemented a custom binary wire format for multiplexing RPC streams over stdin/stdout between parent and privilege-escalated child process, ensuring broad compatibility with sudo implementations and avoiding filesystem pollution from sockets.
- Packaged in major Linux distro repositories, including Arch Linux and Nixpkgs.

astrid.tech

 astrid.tech  May 2020 – Present

Personal website and blog about systems engineering, networking, and low-level software, generated with Rust, Typescript, and Nix. Multiple articles have reached the front page of Hacker News.

Selected writings:

- [curl > /dev/sda¹](https://astrid.tech/2026/03/24/0/curl-to-dev-sda) — a five-part exploration of the Linux boot process, initramfs, Debian and NixOS internals, kexec, and binfmt_misc interpreters.
- [ping ff02::1²](https://astrid.tech/2024/05/28/0/ping-ff02-1) — a practical guide to rescuing systems using IPv6 link-local addresses, and methods for peer-to-peer file transfers without firewall reconfiguration.

Homelab

 github.com/ifd3f/infra ★ 50 stars  Dec 2020 – Jun 2024

Reproducible NixOS-based personal cloud containing a collection of self-hosted services across VPSes and local machines, with S3 buckets provisioned using Terraform.

- Used GitHub Actions CI for build, verification, and deployment of new configurations.
- Previously used k3s Kubernetes for ephemeral service hosting.

Education

B.S. Computer Science — California Polytechnic State University

 San Luis Obispo, CA  Sep 2019 – Aug 2022

- Graduated in 3 years with honors (GPA 3.8).
- *Graduate Computer Graphics (CSC 572)* — Implemented an N-body simulation on GPUs using GLSL compute shaders, capable of simulating a 10,000-star galaxy in real-time.

¹ astrid.tech/2026/03/24/0/curl-to-dev-sda ² astrid.tech/2024/05/28/0/ping-ff02-1